Nicola Novello

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Research Interests

AI safety, machine unlearning, federated unlearning, diffusion models, mutual information estimation and maximization

Education

University of Klagenfurt — *Oct. 2022 – Present*
PhD in Information and Communication Engineering
Supervisors: Prof. Andrea M. Tonello, Prof. Deniz Gündüz (Imperial College London)

University of Padova — *Oct. 2020 – July 2022*
Master's Degree in Control Systems Engineering. Final Mark: 110/110 with honors
Supervisor: Prof. Alessandro Chiuso

University of Padova — *Oct. 2017 – July 2020*
Bachelor's Degree in Information Engineering. Final Mark: 110/110 with honors
Supervisor: Prof. Maria E. Valcher

Work Experience

University of Klagenfurt — *Klagenfurt, Austria*
Teaching Assistant — *Mar. 2023 – Present*
Taught the courses "Machine learning for information and communication engineering" and "Communication systems engineering"

University of Klagenfurt — *Klagenfurt, Austria*
Project Assistant — *Apr. 2025 - Dec. 2025*
Operated research for project "Power Line Communication for Metering - Data Analytics". In collaboration with Kelag International

Princeton University — *Princeton (NJ), US*
Visiting Researcher — *June 2025 - July 2025*
Performed research focused on neural channel capacity estimation. Started collaboration on differentially-private federated unlearning. Under the supervision of Prof. H. Vincent Poor, department of electrical engineering

University of Klagenfurt — *Klagenfurt, Austria*
Project Assistant — *Aug. 2022 - Feb. 2024*
Conducted administrative duties for the European project ERASMUS+ NEON (Network of competence on IoT)

Trumpf — *Vicenza, Italy*
Student Intern — *Mar. 2022 - June 2022*
Developed deep learning algorithm in PyTorch for bending machine trajectory adjustment. Integrated the deep learning algorithm with existing C# infrastructure

Journal Publications

1. **Nicola Novello**, Andrea M. Tonello, Nunzio A. Letizia, Alex Dytso, H. Vincent Poor, "Channel Capacity Estimation with Cooperative Neural Architectures: The Role of Objective Functions and f-Divergences", IEEE Transactions on Communications, *in press*
2. Nunzio A. Letizia, **Nicola Novello**, and Andrea M. Tonello, "Copula Density Neural Estimation", IEEE Transactions on Neural Networks and Deep Learning, 2025
3. **Nicola Novello** and Andrea M. Tonello, "Recurrent DQN for radio fingerprinting with constrained measurements collection", ICT Express, 2025

Conference Publications

1. **Nicola Novello**, Nunzio A. Letizia, Deniz Gündüz, and Andrea M. Tonello, "Insights on f-Divergence-Based Surrogate Objective Functions for Mutual Information Maximization", accepted to Asilomar Conference on Signals, Systems, and Computers 2026
2. **Nicola Novello**, Federico Fontana, Luigi Cinque, Deniz Gündüz, and Andrea M. Tonello, "A Unified Framework for Diffusion Model Unlearning with f-Divergence", ICML 2026
3. **Nicola Novello** and Andrea M. Tonello, "Data-Driven Grid Monitoring Under Label Noise Using Power Line Modems", IEEE SmartGridComm 2025
4. Nunzio A. Letizia, **Nicola Novello**, and Andrea M. Tonello, "Mutual Information Estimation via f-Divergence and Data Derangements", NeurIPS 2024
5. **Nicola Novello** and Andrea M. Tonello, "f-Divergence Based Classification: Beyond the Use of Cross-Entropy", ICML 2024
6. **Nicola Novello** and Andrea M. Tonello, "Label Noise Resistant f-Divergence Receiver for Power Line Communications", IEEE SmartGridComm 2024
7. Frank E. Ebong, **Nicola Novello**, and Andrea M. Tonello, "Human Detection Based on Learning and Classification of Radio Scattering Parameters and Para-Hermitian Eigenvalue Decomposition", IEEE PIMRC 2024
8. **Nicola Novello** and Andrea M. Tonello, "Reinforcement learning for simplified training in fingerprinting radio localization", IEEE BalkanCom 2023

Preprints

1. **Nicola Novello** and Andrea M. Tonello, "Empty SPACE: Cross-Attention Sparsity for Concept Erasure in Diffusion Models", arXiv preprint arXiv:2605.10198. *Submitted to ICLR 2027*
2. **Nicola Novello** and Andrea M. Tonello, "Robust Classification with Noisy Labels Based on Posterior Maximization", arXiv preprint arXiv:2504.06805. *Submitted to IEEE Transactions on Artificial Intelligence (in April 2026)*

Interests and Competences

Programming and software skills: Python, Matlab, C++, Java, LaTeX, GitHub
Expertise: Machine learning, signal processing, information theory
Languages: Italian (native), English (fluent)

Academic Service

Awarded "Gold Reviewer" at ICML 2026.
Reviewer for ICML, ICLR, NeurIPS, AAAI, IEEE Transactions on Communications, IEEE Transactions on Neural Networks and Learning Systems, IEEE VTC